

MSC2050 Discrete Mathematics, Presentation 6

Dr. Anna Tomskova



Inha University in Tashkent

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Divisibility and Modular Arithmetic

(Paragraph 4.1)

Division

If a and b are integers with $a \neq 0$, we say that a divides b if there is an integer c such that $b = ac$.

When a divides b we say that a is a factor or divisor of b , and that b is a multiple of a .

The notation $a \mid b$ denotes that a divides b .

We write $a \nmid b$ when a does not divide b .

11 divides 33, true

15 divides 50, false

33 is a factor of 11, false

5 is a factor of 35, true

4 is a divisor of 20, true

20 is a divisor of 4, false

20 is a multiple of 4, true

7 is a multiple of 21, false

$21 \mid 7$, false

$7 \mid 21$, true

$3 \nmid 6$, false

$6 \nmid 3$, true

Theorem

Let a, b and c be integers, where $a \neq 0$. Then

- (i) if $a \mid b$ and $a \mid c$, then $a \mid (b + c)$;
- (ii) if $a \mid b$, then $a \mid bc$;
- (iii) if $a \mid b$ and $b \mid c$, then $a \mid c$.

Answer Q1

Division Algorithm

Theorem (division algorithm)

Let a be an integer and d a positive integer. Then there are unique integers q and r , with $0 \leq r < d$, such that $a = dq + r$.

In the equality given in the division algorithm, d is called the **divisor**, a is called the **dividend**, q is called the **quotient**, and r is called the **remainder**.

This notation is used to express the quotient and remainder:

$$q = a \mathbf{div} d \quad r = a \mathbf{mod} d.$$

40 **div** 6 is 6

50 **div** 10 is 5

-17 **div** 5 is -4,

-17 = 5(-4) + 3, $0 \leq 3 < 5$.

17 **div** -5 is not defined

5 **div** 7 is 0

5 = 7 * 0 + 5, $0 \leq 5 < 7$.

40 **mod** 6 is 4

50 **mod** 10 is 0

-17 **mod** 5 is 3,

-17 = 5(-4) + 3, $0 \leq 3 < 5$.

17 **mod** -5 is not defined

5 **mod** 7 is 5

5 = 7 * 0 + 5, $0 \leq 5 < 7$.

Answer Q2

Modular Arithmetic

If a and b are integers and m is a positive integer, then a is congruent to b modulo m if m divides $a - b$.

We use the notation $a \equiv b \pmod{m}$

We say that $a \equiv b \pmod{m}$ is a congruence and that m is its modulus (plural moduli).

If a and b are not congruent modulo m , we write $a \not\equiv b \pmod{m}$.

$$20 \equiv 43 \pmod{23} \text{ is true}$$

$$3 \equiv 43 \pmod{23} \text{ is false}$$

$$-20 \equiv 43 \pmod{23} \text{ is false}$$

$$-3 \equiv 43 \pmod{23} \text{ is true}$$

$$3 \equiv -43 \pmod{23} \text{ is true}$$

$$13 \equiv 6 \pmod{7} \text{ is true}$$

$$20 \equiv 6 \pmod{7} \text{ is true}$$

$$27 \equiv 6 \pmod{7} \text{ is true}$$

$$6 \equiv 6 \pmod{7} \text{ is true}$$

$$-1 \equiv 6 \pmod{7} \text{ is true}$$

Theorem

Let a and b be integers, and let m be a positive integer. Then $a \equiv b \pmod{m}$ if and only if $a \bmod m = b \bmod m$.

For example, $20 \equiv 43 \pmod{23}$ means $20 \bmod 23 = 20$ and $43 \bmod 23 = 20$

Answer Q3

Theorem

Let m be a positive integer. The integers a and b are congruent modulo m if and only if there is an integer k such that $a = b + km$.

Proof: If $a \equiv b \pmod{m}$, by the definition of congruence, we know that $m \mid (a - b)$. This means that there is an integer k such that $a - b = km$, so that $a = b + km$. Conversely, if there is an integer k such that $a = b + km$, then $km = a - b$. Hence, m divides $a - b$, so that $a \equiv b \pmod{m}$.

Theorem

Let m be a positive integer. If $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$, then $a + c \equiv b + d \pmod{m}$ and $ac \equiv bd \pmod{m}$.

Proof: Because $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$, there are integers s and t with $b = a + sm$ and $d = c + tm$. Hence,

$$b + d = (a + sm) + (c + tm) = (a + c) + m(s + t)$$

$$bd = (a + sm)(c + tm) = ac + m(at + cs + stm).$$

Hence, $a + c \equiv b + d \pmod{m}$ and $ac \equiv bd \pmod{m}$.

Homework:

- Read “Arithmetic Modulo m ” at p. 243
- Solve *odd* exercises starting at p. 244.

Answer Q4

Integer Representations and Algorithms

(Paragraph 4.2)

Representations of integers

Theorem

Let b be an integer greater than 1. Then if n is a positive integer, it can be expressed **uniquely** in the form

$$n = a_k b^k + a_{k-1} b^{k-1} + \dots + a_1 b + a_0,$$

where k is a nonnegative integer, $a_0, a_1, \dots, a_k$ are nonnegative integers less than b , and $a_k \neq 0$.

The representation of n in the theorem above is called **the base b expansion of n** . The base b expansion of n is denoted by

$$(a_k a_{k-1} \dots a_1 a_0)_b$$

.

e.g. $(245)_8 = 2 \cdot 8^2 + 4 \cdot 8 + 5 = 165$,

where $165 = (165)_{10}$ is **decimal expansion**

How many digits?

Let n be a positive integer.

The base b expansion of n has d digits if

$$b^{d-1} \leq n < b^d.$$

$$d - 1 \leq \log_b n < d$$

$$\lfloor \log_b n \rfloor = d - 1$$

$$d = \lfloor \log_b n \rfloor + 1$$

Important expansions to decimals

Binary expansion is used by computers to represent and do arithmetic with integers.

$$(1\,0101\,1111)_2 = 351$$

Octal expansion:

$$(7016)_8 = 3598$$

Hexadecimal expansion:

$$(2AE0B)_{16} = 175627$$

Decimal	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Hexadecimal	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
Octal	0	1	2	3	4	5	6	7	10	11	12	13	14	15	16	17
Binary	0	1	10	11	100	101	110	111	1000	1001	1010	1011	1100	1101	1110	1111

Conversion from decimals

The octal expansion of 12345 is $(30071)_8$

The hexadecimal expansion of 177130 is $(2B3EA)_{16}$

The binary expansion of 241 is $(1111\,0001)_2$

Constructing base b expansion

procedure base b expansion(n, b : positive integers with $b > 1$)

$q := n$

$k := 0$

while $q \neq 0$

$a_k := q \bmod b$

$q := q \operatorname{div} b$

$k := k + 1$

return $a_{k-1}, \dots, a_1, a_0$

Complexity of this algorithm is $\Theta(\log n)$

Answer Q5

Conversion between binary, octal and hexadecimal

Question: Find octal and hexadecimal expansion of $(11\ 1110\ 1011\ 1100)_2$ and the binary expansions of $(765)_8$ and $(A8D)_{16}$.

Decimal	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Hexadecimal	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
Octal	0	1	2	3	4	5	6	7	10	11	12	13	14	15	16	17
Binary	0	1	10	11	100	101	110	111	1000	1001	1010	1011	1100	1101	1110	1111

Answers: $(37274)_8$, $(3EBC)_{16}$, $(1\ 1111\ 0101)_2$, $(1010\ 1000\ 1101)_2$.

Answer Q6

Addition of two n -bit integers

The sum $(1110)_2 + (1011)_2$ is $(11001)_2$

Algorithm

```
procedure add( $a, b$ : positive integers)
 $\{a = (a_{n-1}a_{n-2} \dots a_1a_0)_2$  and  $b = (b_{n-1}b_{n-2} \dots b_1b_0)_2\}$ 
 $c := 0$ 
for  $j := 0$  to  $n - 1$ 
     $d := \lfloor (a_j + b_j + c)/2 \rfloor$ 
     $s_j := a_j + b_j + c - 2d$ 
     $c := d$ 
 $s_n := c$ 
return  $s_0, s_1, \dots, s_n \{ (s_n s_{n-1} \dots s_0)_2 \}$ 
```

Number of additions is $\Theta(n)$

Multiplication of two n -bit integers

The product $(110)_2 \times (101)_2$ is $(11110)_2$

Algorithm

```
procedure multiply( $a, b$ : positive integers)
{  $a = (a_{n-1}a_{n-2} \dots a_1a_0)_2$  and  $b = (b_{n-1}b_{n-2} \dots b_1b_0)_2$  }
for  $j := 0$  to  $n - 1$ 
    if  $b_j = 1$  then  $c_j := a$  shifted  $j$  places
    else  $c_j := 0$ 
 $p := 0$ 
for  $j := 0$  to  $n - 1$ 
     $p := p + c_j$ 
return  $p$ 
```

Number of shifts is $\Theta(n^2)$

Number of additions is $\Theta(n^2)$

Answer Q7

Modular Exponentiation

In **cryptography** it is important to be able to find $b^n \bmod m$ efficiently, where b , n , and m are large integers.

It is impractical to first compute b^n and then find its remainder when divided by m because b^n will be a huge number.

Instead, we can use the algorithm that employs the binary expansion of the exponent n .

Algorithm

procedure modular exponentiation(b :integer, $n = (a_{k-1}a_{k-2} \dots a_1a_0)_2$,
 m :positive integers)

$x := 1$

$power := b \bmod m$

for $i := 0$ **to** $k - 1$

if $a_i = 1$ **then** $x := (x \cdot power) \bmod m$

$power := (power \cdot power) \bmod m$

return x

Complexity is $O((\log m)^2 \log n)$. For example find $45^{11} \bmod 241$.

Answer Q8

Homework:

- Read “Algorithm for **div** and **mod**” at p. 253
- Solve *odd* exercises starting at p. 255.

Primes and Greatest Common Divisors

(Paragraph 4.3)

Primes

An integer p greater than 1 is called **prime** if the only positive factors of p are 1 and p .

A positive integer that is greater than 1 and is not prime is called **composite**.

The integer n is composite if and only if there exists an integer a such that $a \mid n$ and $1 < a < n$.

7 is prime but 9 is composite since $3 \mid 9$.

THE FUNDAMENTAL THEOREM OF ARITHMETIC

Every integer greater than 1 can be written uniquely as a prime or as the product of two or more primes where the prime factors are written in order of non-decreasing size.

$$100 = 2 \cdot 2 \cdot 5 \cdot 5 = 2^2 5^2,$$

$$641 = 641,$$

$$999 = 3 \cdot 3 \cdot 3 \cdot 37 = 3^3 \cdot 37,$$

$$1024 = 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 = 2^{10}.$$

Trial Division

Theorem

If n is a composite integer, then n has a prime divisor less than or equal to $\sqrt{n}$.

Proof: If n is composite, by the definition of a composite integer, we know that it has a factor a with $1 < a < n$. Hence, by the definition of a factor of a positive integer, we have $n = ab$, where b is a positive integer greater than 1. We will show that $a \leq \sqrt{n}$ or $b \leq \sqrt{n}$. Indeed, if $a > \sqrt{n}$ and $b > \sqrt{n}$, then $ab > \sqrt{n}\sqrt{n} = n$, which is a contradiction. So, $a \leq \sqrt{n}$ or $b \leq \sqrt{n}$. Because both a and b are divisors of n , n has a positive divisor not exceeding $\sqrt{n}$. This divisor is either prime or has a prime divisor less than itself. In either case, n has a prime divisor less than or equal to $\sqrt{n}$.

Corollary

An integer is prime if it is not divisible by any prime less than or equal to its square root.

Show that 101 is prime.

The only primes not exceeding $\sqrt{101}$ are 2, 3, 5, and 7. Because 101 is not divisible by 2, 3, 5, or 7, it follows that 101 is prime.

Answer Q9

Sieve of Eratosthenes

[look at the gif](#)

Theorem

There are infinitely many primes.

Proof: We will prove this theorem using a proof by contradiction.

We assume that there are only finitely many primes, $p_1, p_2, \dots, p_n$.

Let $Q = p_1 p_2 \cdots p_n + 1$.

By the fundamental theorem of arithmetic, Q is prime or else it can be written as the product of two or more primes.

If $p_j \mid Q$, then p_j divides $Q - p_1 p_2 \cdots p_n = 1$.

So none of the primes p_j divides Q .

Hence, there is a prime not in the list $p_1, p_2, \dots, p_n$. This prime is either Q , if it is prime, or a prime factor of Q .

This is a contradiction because we assumed that we have listed all the primes.

Consequently, there are infinitely many primes.

Mersenne primes

There is an ongoing quest to discover larger and larger prime numbers; for almost all the last 300 years, the largest prime known has been an integer of the special form $2^p - 1$, where p is also prime.

Such primes are called **Mersenne primes**, after the French monk Marin Mersenne, who studied them in the seventeenth century.

The reason that the largest known prime has usually been a Mersenne prime is that there is an extremely efficient test, known as the Lucas-Lehmer test, for determining whether $2^p - 1$ is prime.

Furthermore, it is not currently possible to test numbers not of this or certain other special forms anywhere near as quickly to determine whether they are prime. The numbers $2^2 - 1 = 3$, $2^3 - 1 = 7$, $2^5 - 1 = 31$ and $2^7 - 1 = 127$ are Mersenne primes, while $2^{11} - 1 = 2047$ is not a Mersenne prime because $2047 = 23 \cdot 89$. The largest Mersenne prime known is $2^{43112609} - 1$, a number with nearly 13 million decimal digits, which was shown to be prime in 2008.

Known algorithms

- To find **all primes not exceeding** a specified positive integer:
The Sieve of Eratosthenes
with complexity $O(n \log \log n)$
- To determine whether a positive integer **n is prime**:
2002, by Manindra Agrawal, Neeraj Kayal, and Nitin Saxena
with complexity $O((\log n)^6)$
- To find **prime factorization**:
No polynomial-time algorithm for factoring integers is known.
The best known algorithm for big numbers is general number field sieve (GNFS).
For numbers less than 100, Quadratic sieve is better.
They both are of exponential complexity.

Goldbach's Conjecture

- In 1742, Christian Goldbach, in a letter to Leonhard Euler, conjectured that every odd integer n , $n > 5$, is the sum of three primes.
- Euler replied that this conjecture is equivalent to the conjecture that every even integer n , $n > 2$, is the sum of two primes
- The conjecture that every even integer n , $n > 2$, is the sum of two primes is now called **Goldbach's conjecture**.
- For example, $4 = 2 + 2$, $6 = 3 + 3$, $8 = 5 + 3$, $10 = 7 + 3$, $12 = 7 + 5$, and so on.
- In 2011, the conjecture has been checked for all positive even integers up to $1.6 \cdot 10^{18}$.
- No proof of Goldbach's conjecture has been found!

Greatest Common Divisors

Let a and b be integers, not both zero. The largest integer d such that $d \mid a$ and $d \mid b$ is called the **greatest common divisor of a and b** .

The greatest common divisor of a and b is denoted by $\gcd(a, b)$.

$\gcd(24, 36) = 12$, the common divisors of 24 and 36 are 1, 2, 3, 4, 6, and 12.

$\gcd(17, 22) = 1$, 17 and 22 have no positive common divisors other than 1.

The integers a and b are **relatively prime** if their greatest common divisor is 1.

The integers $a_1, a_2, \dots, a_n$ are **pairwise relatively prime** if $\gcd(a_i, a_j) = 1$.

10, 17, and 21 are pairwise relatively prime

10, 19, and 24 are not pairwise relatively prime.

Theorem

If $a = p_1^{a_1} p_2^{a_2} \dots p_n^{a_n}$ and $b = p_1^{b_1} p_2^{b_2} \dots p_n^{b_n}$ then

$$\gcd(a, b) = p_1^{\min(a_1, b_1)} p_2^{\min(a_2, b_2)} \dots p_n^{\min(a_n, b_n)}$$

$\gcd(150, 500) = 20$.

Answer Q10

Least common multiple

The **least common multiple** of the positive integers a and b is the smallest positive integer that is divisible by both a and b .

The least common multiple of a and b is denoted by $\text{lcm}(a, b)$.

Theorem

If $a = p_1^{a_1} p_2^{a_2} \dots p_n^{a_n}$ and $b = p_1^{b_1} p_2^{b_2} \dots p_n^{b_n}$ then

$$\text{lcm}(a, b) = p_1^{\max(a_1, b_1)} p_2^{\max(a_2, b_2)} \dots p_n^{\max(a_n, b_n)}$$

$$\text{lcm}(15, 24) = 120.$$

Theorem

Let a and b be positive integers. Then

$$ab = \text{gcd}(a, b) \cdot \text{lcm}(a, b).$$

The Euclidean Algorithm

The Euclidean algorithm is a method of finding the gcd.

Algorithm

procedure gcd(a, b : positive integers)

$x := a$

$y := b$

while $y \neq 0$

$r := x \bmod y$

$x := y$

$y := r$

return x

gcd(91, 287) = 7.

Lemma

Let $a = bq + r$, where a, b, q and r are integers. Then $\text{gcd}(a, b) = \text{gcd}(b, r)$.

Complexity of the algorithm above is $O(\log b)$

gcd as linear combination

Bézout's theorem

If a and b are positive integers, then there exist integers s and t such that

$$\gcd(a, b) = sa + tb.$$

The integers s and t above are called **Bézout coefficients** of a and b . Also, the equation $\gcd(a, b) = sa + tb$ is called **Bézout's identity**.

Extended Euclidean algorithm

Express $\gcd(252, 198) = 18$ as a linear combination of 252 and 198.

The Euclidean algorithm is

$$252 = 1 \cdot 198 + 54$$

$$198 = 3 \cdot 54 + 36$$

$$54 = 1 \cdot 36 + 18$$

$$36 = 2 \cdot 18.$$

$$\begin{aligned} 18 &= 54 - 1 \cdot 36 = 54 - 1 \cdot (198 - 3 \cdot 54) = 4 \cdot 54 - 1 \cdot 198 = \\ &= 4 \cdot (252 - 1 \cdot 198) - 1 \cdot 198 = 4 \cdot 252 - 5 \cdot 198. \end{aligned}$$

Answer Q11

Lemma 1

If a, b and c are positive integers such that $\gcd(a, b) = 1$ and $a \mid bc$, then $a \mid c$.

Proof: Because $\gcd(a, b) = 1$, by Bézout's theorem there are integers s and t such that $sa + tb = 1$.

Multiplying both sides of this equation by c , we obtain $sac + tbc = c$.

By Theorem on slide 3, part (ii), $a \mid tbc$.

Because $a \mid sac$ and $a \mid tbc$, by (i) of that theorem, we conclude that a divides $c = sac + tbc$.

Lemma 2

If p is a prime and $p \mid a_1 a_2 \dots a_n$, where each a_i is an integer, then $p \mid a_i$ for some i .

Proof is left as an exercise.

Proof (of the uniqueness of the prime factorization of a positive integer):

Suppose that the positive integer n can be written as the product of primes in two different ways, say,

$$n = p_1 p_2 \dots p_s \quad \text{and} \quad n = q_1 q_2 \dots q_t,$$

each p_i and q_j are primes such that

$$p_1 \leq p_2 \leq \dots \leq p_s \quad \text{and} \quad q_1 \leq q_2 \leq \dots \leq q_t.$$

When we remove all common primes from the two factorizations, we have

$$p_{i_1} p_{i_2} \dots p_{i_u} = q_{j_1} q_{j_2} \dots q_{j_v},$$

where no prime occurs on both sides of this equation.

By Lemma 2 it follows that p_{i_1} divides q_{j_k} for some k .

Because no prime divides another prime, this is impossible.

Division of congruence

The congruence $14 \equiv 8 \pmod{6}$ holds, but both sides of this congruence cannot be divided by 2 to produce a valid congruence $7 \not\equiv 4 \pmod{6}$.

Theorem

Let m be a positive integer and let a, b and c be integers. If $ac \equiv bc \pmod{m}$ and $\gcd(c, m) = 1$, then $a \equiv b \pmod{m}$.

Proof: Because $ac \equiv bc \pmod{m}$, $m \mid ac - bc = c(a - b)$.

By Lemma 1, because $\gcd(c, m) = 1$, it follows that $m \mid a - b$.

We conclude that $a \equiv b \pmod{m}$.

For example, $14 \equiv 20 \pmod{3}$ and $7 \equiv 10 \pmod{3}$.

Homework:

- Solve *odd* exercises starting at p. 272.